OS LAB

NAME : GEETHA S

USN: 1RUA25BCA0033

1. **Process Creation in Unix Shell: Creating Child Processes and Displaying Parent-Child Relationships Using ps.**

```
geetha@Geetha:~$ cd os
geetha@Geetha:~/os$ nano prog1.sh
geetha@Geetha:~/os$ chmod +x prog1.sh
geetha@Geetha:~/os$ ./prog1.sh
Parent PID: 449
“Child PID: 450”
    PID    PPID CMD
    302     301 -bash
    449     302 -bash
    450     449 sleep 20
    451     449 ps -o pid,ppid,cmd
geetha@Geetha:~/os$ nano prog1.sh
geetha@Geetha:~/os$ chmod +x prog1.sh
geetha@Geetha:~/os$ ./prog1.sh
Parent PID: 491
“Child PID: 492”
geetha@Geetha:~/os$ ps -o pid,ppid,cmd
    PID    PPID CMD
    302     301 -bash
    495     302 ps -o pid,ppid,cmd
geetha@Geetha:~/os$ sleep 30
^Z
[1]+  Stopped                 sleep 30
geetha@Geetha:~/os$ sleep 50
^Z
[2]+  Stopped                 sleep 50
geetha@Geetha:~/os$ pstree -p $$
bash(302)───pstree(506)
           ├─sleep(496)
           └─sleep(499)
geetha@Geetha:~/os$ pstree -p
systemd(1)─┬─agetty(197)
           ├─agetty(203)
           ├─cron(168)
           ├─dbus-daemon(169)
           ├─init-systemd(Ub(2)─┬─SessionLeader(300)───Relay(302)(301)───ba+
                                ├─init(6)───{init}(7)
                                ├─login(303)───bash(373)
                                └─{init-systemd(Ub}(8)
           ├─rsyslogd(201)─┬─{rsyslogd}(219)
                           ├─{rsyslogd}(220)
                           └─{rsyslogd}(221)
           ├─systemd(348)───(sd-pam)(349)
```

2. **Process Control: Create a shell script that demonstrates process control commands like bg, fg, jobs, and kill. Implement a simple background task and manage its execution.**

```
            ├─systemd(348)───(sd-pam)(349)
            ├─systemd-journal(42)
            ├─systemd-logind(177)
            ├─systemd-resolve(107)
            ├─systemd-timesyn(108)───{systemd-timesyn}(157)
            ├─systemd-udevd(92)
            ├─unattended-upgr(209)───{unattended-upgr}(256)
            └─wsl-pro-service(181)─┬─{wsl-pro-service}(212)
                                   ├─{wsl-pro-service}(213)
                                   ├─{wsl-pro-service}(214)
                                   ├─{wsl-pro-service}(215)
                                   ├─{wsl-pro-service}(218)
                                   ├─{wsl-pro-service}(225)
                                   └─{wsl-pro-service}(228)
geetha@Geetha:~/os$ jobs
[1]-  Stopped                 sleep 30
[2]+  Stopped                 sleep 50
geetha@Geetha:~/os$ sleep 500
^Z
[3]+  Stopped                 sleep 500
geetha@Geetha:~/os$ sleep 300
^Z
[4]+  Stopped                 sleep 300
geetha@Geetha:~/os$ cat
^Z
[5]+  Stopped                 cat
geetha@Geetha:~/os$ jobs
[1]   Stopped                 sleep 30
[2]   Stopped                 sleep 50
[3]   Stopped                 sleep 500
[4]-  Stopped                 sleep 300
[5]+  Stopped                 cat
geetha@Geetha:~/os$ bg %2
[2] sleep 50 &
geetha@Geetha:~/os$ jobs
[1]   Stopped                 sleep 30
[2]   Done                    sleep 50
[3]   Stopped                 sleep 500
[4]-  Stopped                 sleep 300
[5]+  Stopped                 cat
geetha@Geetha:~/os$ fg %3
sleep 500
^Z
```

3. **Basic Shell Script for Job Scheduling: Develop a shell script that takes a list of commands and executes them with a specified delay using the sleep command.Demonstrate job scheduling in the shell.**

Output

```
geetha@Geetha:~/os$ nano prog3.sh
geetha@Geetha:~/os$ chmod +x prog3.sh
geetha@Geetha:~/os$ ./prog3.sh
job 1: Display Date
Tue Apr  7 14:49:44 UTC 2026
job 2: Display current user
geetha
job 3: Display Calender
./prog3.sh: line 10: cal: command not found
 job 4: list the files
 error.txt   lab1.sh   lab7.sh   lab9.sh   prog1.sh
 file1.txt   lab6.sh   lab8.sh  'os lab'   prog3.sh
job 5: display my system date
14:49:55
job 6: print working directory
/home/geetha/os
All jobs are finished
geetha@Geetha:~/os$ sudo apt update
[sudo] password for geetha:

[6]+  Stopped                 sudo apt update
```

4. **File Operations: Create a shell script that performs file operations, such as creating, copying, moving, and deleting files using commands like touch, cp, mv, and rm. Include error handling to manage file operations gracefully. In shell scripting (file operations), -f and -e are file test operators used with if**
   <u>CONDITION</u>
   - **-f → Checks if file exists AND is a regular file - if [-f filename] ⍰ normal file**
   - **-e → Checks if file exists (any type) - if [-e filename] ⍰ File exists (file / directory / link anything)**

```
  GNU nano 7.2                                                    prog4.sh
#!/bin/bash

while true
do
echo "-------------- FILE OPERATIONS ----------"
echo "1. Create File"
echo "2. Copy File"
echo "3. Move File"
echo "4. Delete File"
echo "5. Display File"
echo "6. Exit"

echo -n "Enter your choice: "
read choice

if [ $choice -eq 1 ]; then
    read -p "Enter filename to create: " fname
    if [ -f "$fname" ]; then
        echo "Error: File already exists!"
    else
        read -p "Enter content: " content
        echo "$content" > "$fname"
        echo "File '$fname' created successfully."
    fi

elif [ $choice -eq 2 ]; then
    read -p "Enter source filename: " source
    read -p "Enter destination filename: " dest
    if [ -f "$source" ]; then
        cp "$source" "$dest"
        echo "File copied successfully."
    else
        echo "Error: Source file does not exist!"
    fi

elif [ $choice -eq 3 ]; then
```

```
    if [ -f "$source" ]; then
        mv "$source" "$dest"
        echo "File moved successfully."
    else
        echo "Error: File does not exist!"
    fi

elif [ $choice -eq 4 ]; then
    read -p "Enter filename to delete: " fname
    if [ -f "$fname" ]; then
        rm "$fname"
        echo "File deleted successfully."
    else
        echo "Error: File does not exist!"
    fi

elif [ $choice -eq 5 ]; then
    read -p "Enter filename to display: " fname
    if [ -f "$fname" ]; then
        echo "Content of '$fname':"
        cat "$fname"
    else
        echo "Error: File does not exist!"
    fi

elif [ $choice -eq 6 ]; then
    echo "Exiting program..."
    break

else
    echo "Invalid choice!"
fi

echo ""
done
```

OUTPUT

```
geetha@Geetha:~/os$ nano lab4.sh
geetha@Geetha:~/os$ chmod +x lab4.sh
geetha@Geetha:~/os$ ./lab4.sh
 -------------- FILE OPERATIONS ----------
1. Create File
2. Copy File
3. Move File
4. Delete File
5. Display File
6. Exit
Enter your choice: 1
Enter filename to create: Geetha
Enter content: I am newly created file
File 'Geetha' created successfully.

 -------------- FILE OPERATIONS ----------
1. Create File
2. Copy File
3. Move File
4. Delete File
5. Display File
6. Exit
Enter your choice: 6
Exiting program...
```

```
-------------- FILE OPERATIONS ----------
1. Create File
2. Copy File
3. Move File
4. Delete File
5. Display File
6. Exit
Enter your choice: 3
Enter filename to move: file5.txt
Enter new filename: newfile.txt
File moved successfully.

-------------- FILE OPERATIONS ----------
1. Create File
2. Copy File
3. Move File
4. Delete File
5. Display File
6. Exit
Enter your choice: 4
Enter filename to delete: file4.txt
File deleted successfully.

-------------- FILE OPERATIONS ----------
1. Create File
2. Copy File
3. Move File
4. Delete File
5. Display File
6. Exit
Enter your choice: 6
Exiting program...
```

5. **File Permissions: Write a shell script to modify file permissions and ownershig using chmod and chown. Include user input for the filename and desired permissions and display the new file permissions.**

```
  GNU nano 7.2                                                    prog5.sh
echo "---- File Permission & Ownership ----"
echo "Enter file name:"
read file
if [ ! -f "$file" ]; then
echo "File not found!"
exit
fi
echo "Enter permission (e.g., 755):"
read perm
chmod $perm $file
echo "Enter new owner name:"
read owner
chown $owner $file
echo "Updated file details:"
ls -l $file
```

OUTPUT

```
(base) rvu@rvu-OptiPlex-SFF-7020:~/Geetha$ nano lab5.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/Geetha$ ./lab5.sh
-----File Permission & Ownership-----
Enter file name:
lab5.sh
Enter permission (e.g.,755):
788
chmod: missing operand after ‘lab5.sh’
Try 'chmod --help' for more information.
Enter new owner name
harshitha
chmod: invalid mode: ‘harshitha’
Try 'chmod --help' for more information.
Updated file details
-rwxrwxr-x 1 rvu rvu 319 Mar 27 12:47 lab5.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/Geetha$
```

6. Directory Management: Develop a shell script that creates, lists, and deletes directories. Use commands like mkdir, ls, and rmdir, and include user prompts for directory names.

```
echo "Lab 6: Directory Management"
while true
do
    echo ""
    echo "1. Create Directory"
    echo "2. List Directories"
    echo "3. Delete Directory"
    echo "4. Exit"
    echo "Enter your choice:"
    read choice
    case $choice in
    1)
        echo "Enter directory name to create:"
        read dirname
        mkdir "$dirname"
        if [ $? -eq 0 ]; then
            echo "Directory '$dirname' created successfully."
        else
            echo "Error creating directory."
        fi
        ;;
    2)
        echo "Directories in current location:"
        ls -d */
        ;;
    3)
        echo "Enter directory name to delete:"
        read dirname
        rmdir "$dirname"
        if [ $? -eq 0 ]; then
            echo "Directory '$dirname' deleted successfully."
        else
            echo "Error deleting directory (may not be empty)."
        fi
        ;;
    4)
```

```
        fi
        ;;
    4)
        echo "Exiting program..."
        break
        ;;
    *)
        echo "Invalid choice. Try again!"
        ;;
    esac
done
```

OUTPUT

```
geetha@Geetha:~$ mkdir os
geetha@Geetha:~$ cd os
geetha@Geetha:~/os$ nano lab6.sh
geetha@Geetha:~/os$ chmod +x lab6.sh
geetha@Geetha:~/os$ ./lab6.sh
Lab 6: Directory Management

1. Create Directory
2. List Directories
3. Delete Directory
4. Exit
Enter your choice:
1
Enter directory name to create:
os lab
Directory 'os lab' created successfully.

1. Create Directory
2. List Directories
3. Delete Directory
4. Exit
Enter your choice:
4
Exiting program...
geetha@Geetha:~/os$ nano lab7.sh
geetha@Geetha:~/os$ chmod +x lab7.sh
geetha@Geetha:~/os$ ./lab7.sh
Lab 7: Memory Usage Monitor

Current Memory Status (using free command):
               total        used        free      shared  buff/cache   available
Mem:           7.6Gi       509Mi       7.1Gi       3.5Mi       204Mi       7.1Gi
Swap:          2.0Gi          0B       2.0Gi

Detailed Memory Info (using top command):
MiB Mem :   7785.2 total,   7219.8 free,    510.6 used,    205.0 buff/cache

Available Memory:
Available: 7.1Gi
geetha@Geetha:~/os$ nano lab8.sh
geetha@Geetha:~/os$ chmod +x lab8.sh
geetha@Geetha:~/os$ ./lab8.sh
Lab 8: Environment Variables Demo

Local Variables:
```

7. Dynamic Memory Check: Write a shell script that monitors memory usage on the system using commands like free and top. Display the current memory usage and available memory.

Memory usage → shows how much RAM is being used by the system

Free memory → available RAM for new processes

free command → gives summary of total, used, and free memory

top command → shows real-time system usage (CPU + memory)

-h option in free → displays values in human-readable format (MB/GB)

```
  GNU nano 7.2                                                    lab7.sh
echo "Lab 7: Memory Usage Monitor"
echo ""
echo "Current Memory Status (using free command):"
free -h
echo ""
echo "Detailed Memory Info (using top command):"
top -b -n 1 | grep "MiB Mem"
echo ""
echo "Available Memory:"
free -h | grep "Mem" | awk '{print "Available: " $7}'
```

OUTPUT

```
geetha@Geetha:~$ mkdir os
geetha@Geetha:~$ cd os
geetha@Geetha:~/os$ nano lab6.sh
geetha@Geetha:~/os$ chmod +x lab6.sh
geetha@Geetha:~/os$ ./lab6.sh
Lab 6: Directory Management

1. Create Directory
2. List Directories
3. Delete Directory
4. Exit
Enter your choice:
1
Enter directory name to create:
os lab
Directory 'os lab' created successfully.

1. Create Directory
2. List Directories
3. Delete Directory
4. Exit
Enter your choice:
4
Exiting program...
geetha@Geetha:~/os$ nano lab7.sh
geetha@Geetha:~/os$ chmod +x lab7.sh
geetha@Geetha:~/os$ ./lab7.sh
Lab 7: Memory Usage Monitor

Current Memory Status (using free command):
               total        used        free      shared  buff/cache   available
Mem:           7.6Gi       509Mi       7.1Gi       3.5Mi       204Mi       7.1Gi
Swap:          2.0Gi          0B       2.0Gi

Detailed Memory Info (using top command):
MiB Mem :   7785.2 total,   7219.8 free,    510.6 used,    205.0 buff/cache

Available Memory:
Available: 7.1Gi
geetha@Geetha:~/os$ nano lab8.sh
geetha@Geetha:~/os$ chmod +x lab8.sh
geetha@Geetha:~/os$ ./lab8.sh
Lab 8: Environment Variables Demo

Local Variables:
```

8. Environment Variables: Create a shell script that sets, exports, and displays environment variables. Use printenv and env to show the current environment settings.

```bash
#!/bin/bash

echo "Lab 8: Environment Variables Demo"

# Setting variable
name="BHAVYA SAHITHI G"
course="Operating Systems"

echo ""
echo "Local Variables:"
echo "Name: $name"
echo "Course: $course"

# Exporting variables
export name
export course

echo ""
echo "Using printenv command:"
printenv name
printenv course

echo ""
echo "All Environment Variables (using env):"
env
```

OUTPUT

```
geetha@Geetha:~$ mkdir os
geetha@Geetha:~$ cd os
geetha@Geetha:~/os$ nano lab6.sh
geetha@Geetha:~/os$ chmod +x lab6.sh
geetha@Geetha:~/os$ ./lab6.sh
Lab 6: Directory Management

1. Create Directory
2. List Directories
3. Delete Directory
4. Exit
Enter your choice:
1
Enter directory name to create:
os lab
Directory 'os lab' created successfully.

1. Create Directory
2. List Directories
3. Delete Directory
4. Exit
Enter your choice:
4
Exiting program...
geetha@Geetha:~/os$ nano lab7.sh
geetha@Geetha:~/os$ chmod +x lab7.sh
geetha@Geetha:~/os$ ./lab7.sh
Lab 7: Memory Usage Monitor

Current Memory Status (using free command):
               total        used        free      shared  buff/cache   available
Mem:           7.6Gi       509Mi       7.1Gi       3.5Mi       204Mi       7.1Gi
Swap:          2.0Gi          0B       2.0Gi

Detailed Memory Info (using top command):
MiB Mem :   7785.2 total,   7219.8 free,    510.6 used,    205.0 buff/cache

Available Memory:
Available: 7.1Gi
geetha@Geetha:~/os$ nano lab8.sh
geetha@Geetha:~/os$ chmod +x lab8.sh
geetha@Geetha:~/os$ ./lab8.sh
Lab 8: Environment Variables Demo

Local Variables:
```

```
Local Variables:
Name: Thenmozhi
Course: Operating Systems

Using printenv command:
Thenmozhi
Operating Systems

All Environment Variables (using env):
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
WT_SESSION=7304976e-d328-4004-a214-ec7a6f000b91
NAME=Geetha
PWD=/home/geetha/os
LOGNAME=geetha
HOME=/home/geetha
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/723_interop
WAYLAND_DISPLAY=wayland-0
TERM=xterm-256color
USER=geetha
DISPLAY=:0
SHLVL=2
XDG_RUNTIME_DIR=/run/user/1000/
WSLENV=WT_SESSION:WT_PROFILE_ID:
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Wi
ndows/System32/WindowsPowerShell/v1.0/:/mnt/c/Windows/System32/OpenSSH/:/mnt/c/Program Files/dotnet/:/mnt/c/Program Files/Git/cmd:/mnt/c/Users/Dell/AppData/Local/Programs/Python/Python314/
Scripts/:/mnt/c/Users/Dell/AppData/Local/Programs/Python/Python314/:/mnt/c/Users/Dell/AppData/Local/Programs/Python/Launcher/:/mnt/c/Users/Dell/AppData/Local/Microsoft/WindowsApps:/snap/bi
n
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
course=Operating Systems
PULSE_SERVER=unix:/mnt/wslg/PulseServer
WT_PROFILE_ID={3c2d458e-0403-5c08-aedd-909080ea585e}
OLDPWD=/home/geetha
name=Thenmozhi
_=/usr/bin/env
geetha@Geetha:~/os$ nano lab8.sh
geetha@Geetha:~/os$ chmod +x lab8.sh
geetha@Geetha:~/os$ ./lab8.sh
Lab 8: Environment Variables Demo
```

9. I/O Redirection in Unix Shell: Manipulating Data Streams with Enhanced File Operations.

```
  GNU nano 7.2                                                    lab9.sh
#!/bin/bash

echo "Lab 9: I/O Redirection Demo"

# Output redirection (overwrite)
echo "Hello, this is OS Lab" > file1.txt
echo "Content written to file1.txt"

# Append output
echo "Appending new line" >> file1.txt
echo "Content appended to file1.txt"

# Input redirection
echo ""
echo "Reading from file1.txt using input redirection:"
cat < file1.txt

# Error redirection
echo ""
echo "Trying to read non-existing file:"
cat file_not_exist.txt 2> error.txt
echo "Error message stored in error.txt"

# Display error file
echo ""
echo "Error file content:"
cat error.txt
```

OUTPUT

```
Using printenv command:
Geetha
Operating Systems

All Environment Variables (using env):
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
WT_SESSION=7304976e-d328-4004-a214-ec7a6f000b91
NAME=Geetha
PWD=/home/geetha/os
LOGNAME=geetha
HOME=/home/geetha
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/723_interop
WAYLAND_DISPLAY=wayland-0
TERM=xterm-256color
USER=geetha
DISPLAY=:0
SHLVL=2
XDG_RUNTIME_DIR=/run/user/1000/
WSLENV=WT_SESSION:WT_PROFILE_ID:
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Wi
ndows/System32/WindowsPowerShell/v1.0/:/mnt/c/Windows/System32/OpenSSH/:/mnt/c/Program Files/dotnet/:/mnt/c/Program Files/Git/cmd:/mnt/c/Users/Dell/AppData/Local/Programs/Python/Python314/
Scripts/:/mnt/c/Users/Dell/AppData/Local/Programs/Python/Python314/:/mnt/c/Users/Dell/AppData/Local/Programs/Python/Launcher/:/mnt/c/Users/Dell/AppData/Local/Microsoft/WindowsApps:/snap/bi
n
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
course=Operating Systems
PULSE_SERVER=unix:/mnt/wslg/PulseServer
WT_PROFILE_ID={3c2d458e-0403-5c08-aedd-909080ea585e}
OLDPWD=/home/geetha
name=Geetha
_=/usr/bin/env
geetha@Geetha:~/os$ nano lab9.sh
geetha@Geetha:~/os$ chmod +x lab9.sh
geetha@Geetha:~/os$ ./lab9.sh
Lab 9: I/O Redirection Demo
Content written to file1.txt
Content appended to file1.txt

Reading from file1.txt using input redirection:
Hello, this is OS Lab
Appending new line
```

10. Command Piping in Unix Shell: Exploring Operating System Features.

```bash
#!/bin/bash

echo "Command Piping Demo"

echo -e "apple\nbanana\napple\norange\nbanana\napple" > fruits.txt

echo "Sample file 'fruits.txt' created with content:"
cat fruits.txt
echo

total_lines=$(cat fruits.txt | wc -l)
echo "Total lines in fruits.txt: $total_lines"

apple_count=$(cat fruits.txt | grep "apple" | wc -l)
echo "Number of times 'apple' appears: $apple_count"

echo "Sorted unique fruits:"
cat fruits.txt | sort | uniq
echo

echo "Lines containing 'banana':"
cat fruits.txt | grep "banana"

rm fruits.txt
echo
echo "Demo complete. Sample file removed."
```

OUTPUT

```
geetha@Geetha:~/os$ nano lab10.sh
geetha@Geetha:~/os$ chmod +x lab10.sh
geetha@Geetha:~/os$ ./lab10.sh
Command Piping Demo
Sample file 'fruits.txt' created with content:
apple
banana
apple
orange
banana
apple

Total lines in fruits.txt: 6
Number of times 'apple' appears: 3
Sorted unique fruits:
apple
banana
orange

Lines containing 'banana':
banana
banana

Demo complete. Sample file removed.
geetha@Geetha:~/os$
```